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Bihar Engineering University, Patna

B.Tech 5th Semester Exam., 2024

Code: 101502 ***Time: 03 Hours***

Semester: V ***Subject: ENVIRONMENTAL ENGINEERING-I*** ***Full Marks: 70***

Instructions:

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) As per IS:1172-1963, water required per head per day for average domestic purposes is approximately equal to

(i) 125 liters (iii) 75 liters
(ii) 135 liters (iv) 350 liters

(b) A water sample with a high biochemical oxygen demand (BOD) and low dissolved oxygen (DO) indicates:

(i) High organic pollution (iii) Good water quality
(ii) Highly chlorinated water (iv) Hard water

(c) When source of pervious strata, the surface of water surrounding the well is at atmospheric pressure, the well is known as

(i) Gravity well (iii) Artesian Well
(ii) Open well (iv) Deep well

(d) The critical settling velocity in a sedimentation tank is calculated based on:

(i) Flow rate and tank width (iii) Surface area and flow rate
(ii) Volume and detention time (iv) Water temperature

(e) The effectiveness of UV disinfection decreases significantly when:

(i) pH increases above 8 (iii) Turbidity exceeds 5 NTU
(ii) Temperature drops below 20°C (iv) Alkalinity is too high

(f) According to WHO, what is the permissible chloride content in drinking water?

(i) .001 mg/l (iii) 1 mg/l
(ii) 200 mg/l (iv) 350 mg/l

(g) As per IS: 4954-1968, the acceptable outdoor noise level for residential and business urban area is

(i) 30-40 dB (iii) 40-50 dB
(ii) 35-45 dB (iv) 50-60 dB

(h) An automobile emits CO = 2.5 g/km. If a car runs for 80 km/day, calculate total CO emitted in a week.

(i) 1.2 kg (iii) 1.6 kg
(ii) 1.4 kg (iv) 1.8 kg

(i) What is the most common secondary pollutant in automobile emissions?

(i) CO (iii) NO
(ii) PAN (iv) SO_2

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(j) A dBA scale is:

(i) Logarithmic weighted scale for human hearing
(ii) Scale used only for machines
(iii) Absolute sound level
(iv) Digital acoustic balance

Q.2 Solve both questions : **[7 x 2 = 14]**

(a) Explain the classification of water sources based on quality and reliability.

(b) Calculate the per capita demand if a town of 50,000 people consumes 10 million liters of water daily.

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) A diagram of a rapid sand filter is shown. Label all parts and explain the filtration process.

(b) A rapid sand filter has a filtration rate of 5 $m^3/m^2/hr$ and an area of 60 m^2. Calculate the total volume of water filtered in 10 hours.

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) Explain any three methods of estimating the future population of a city. What are their relative merits?

(b) A city has following recorded population:

Year	Population
1951	50000
1971	110000
1991	160000

Estimate: (i) the saturation population
(ii) expected population in 2011

Q.5 Solve both questions : **[7 x 2 = 14]**

(a) Define the major components of atmospheric air and their typical percentage composition.

(b) Calculate the mass of nitrogen present in 10 m^3 of air at STP (Assume air is 78% nitrogen by volume, density of air ≈ 1.29 kg/m^3).

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Discuss the selection criteria for plumbing fixtures in multi-storey buildings.

(b) Differentiate between water supply and waste water plumbing in domestic buildings.

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) Evaluate the role of venting systems in maintaining hygiene and preventing trap seal loss in building drainage.

(b) Study the drainage system layout shown. Label any three components and explain how improper venting can impact them.

Q.8 Solve both questions : **[7 x 2 = 14]**

(a) Discuss the chemical and physical principles involved in coagulation and flocculation. How does pH affect coagulant performance?

(b) Evaluate the factors affecting the efficiency of a sedimentation tank. What are the limitations of plain sedimentation for turbidity removal?

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Q.9 Solve both questions : **[7 x 2 = 14]**

(a) Explain the IS 10500 recommendations for residual chlorine in drinking water. Why is maintaining residual chlorine critical even after primary treatment?

(b) If a contact tank is required to maintain 0.2 mg/L of free residual chlorine for 30 minutes, determine the chlorine dose required for a water flow of 10 MLD assuming chlorine demand is 1.2 mg/L.

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